ALICE L WALTON SCHOOL OF MEDICINE

Carotid Stenosis

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Adapted from: Carotid Stenosis

Listen to this Brick
0:00 / 22:45 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of carotid stenosis.
2. Explain the pathophysiology of carotid stenosis.
3. Interpret clinical findings to properly diagnose carotid stenosis.
4. Outline the management plan of carotid stenosis.

CASE CONNECTION

"All I said was that I couldn't see out of my right eye for a few minutes. Then they sent me for these tests, and now they're talking about doing surgery!" This begins your conversation with TR, a 62-year-old female who is seeking a second opinion. She has hypertension, diabetes mellitus, and hypercholesterolemia and smokes a pack of cigarettes per day. She tells you that while playing the piano 3 days ago, she couldn't see the notes on the upper part of her sheet music with her left eye closed. "It felt like someone was closing a window shade inside my right eye. After a few minutes, everything was fine. It may have happened 1 week before this, but, both times, it all went away." On exam, TR's blood pressure is 142/92 mm Hg. She has a faint right carotid bruit. Her neurologic exam is normal. On a copy of the carotid ultrasound and CT angiography she brought with her, 95% stenosis of the right carotid artery is noted.

What is the likely diagnosis? What advice would you give TR? Consider these questions as you read, and we'll revisit TR at the end at the end of the brick.

GO TO CONCLUSION ↓

What Is Carotid Stenosis?

The two carotid arteries—one on each side of the neck—carry oxygenated blood from the heart to the brain. In carotid stenosis, these arteries are narrowed, usually from atherosclerosis. This narrowing limits blood flow to the brain, but can also lead to **emboli** that dislodge and move to smaller arteries in the brain, causing stroke or ischemia and leading to neurologic symptoms and signs.

How Do Patients With Carotid Stenosis Present? (LO #1)

Early in its course, carotid stenosis is asymptomatic. The narrowing itself does not usually cause symptoms of low blood flow to the brain (e.g., headache, fatigue, dizziness, or syncope) except in severe bilateral disease.

Generally, the only "symptom" of carotid artery disease is a complication, such as a stroke or transient ischemic attack (TIA), usually resulting from embolization of a clot from the carotid thrombosis to the brain.

How does carotid stenosis typically present?

What Is the Pathophysiology of Carotid Stenosis? (LO #2)

Carotid stenosis is pr[illegible]igure 1). The atherosclerotic plaque narrows the artery, reducing blood flow. Importantly, the narrowing can also lead to the formation of a carotid thrombus (a clot that forms at a site of endothelial injury), which may give rise to an embolus (a piece of clot released into the bloodstream after a plaque ruptures, which travels as far as the vessel's diameter allows). Emboli are crucial to the consequences of carotid artery disease, because the main mechanism of stroke occurs when an embolus from the ruptured plaque travels downstream until it occludes smaller branches of the carotid artery such as the central retinal or middle cerebral artery. Such a process usually results in oxygenation deficits and may present as a neurologic deficit, either transient/reversible (as in a **TIA**) or permanent (as in **stroke**).

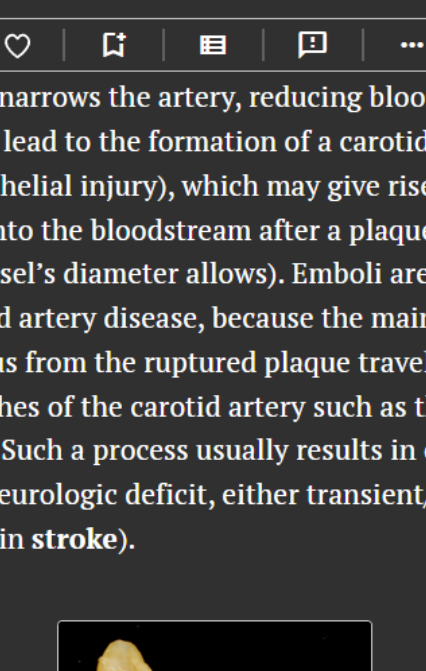

Figure 1. Carotid Artery Atherosclerosis. Note the presence of a large atherosclerotic plaque at the bifurcation of the common carotid artery.

The narrowing of the artery itself is unlikely to cause symptoms because of collateral circulation. With less than 70% stenosis, the contralateral carotid and vertebral arteries (via the circle of Willis) can compensate for the reduced flow through the occluded artery.

Note that symptoms from low flow can sometimes occur with a mild-to-moderate degree of carotid stenosis if the patient also has another cause of low blood flow, such as heart failure or volume depletion.

Why does narrowing of the carotid arteries usually cause no symptoms?

Complications

The main consequences of carotid artery disease are from its complications, including stroke, TIA, and retinal artery occlusion.

Stroke. Carotid stenosis increases the risk of stroke. An asymptomatic person whose carotid artery is more than 50% stenosed has a 0.5% to 1% annual risk of stroke.

Embolic strokes are most common, in which an embolus breaks off, travels upstream, and lodges in an artery supplying part of the brain (Figures 2 and 3).

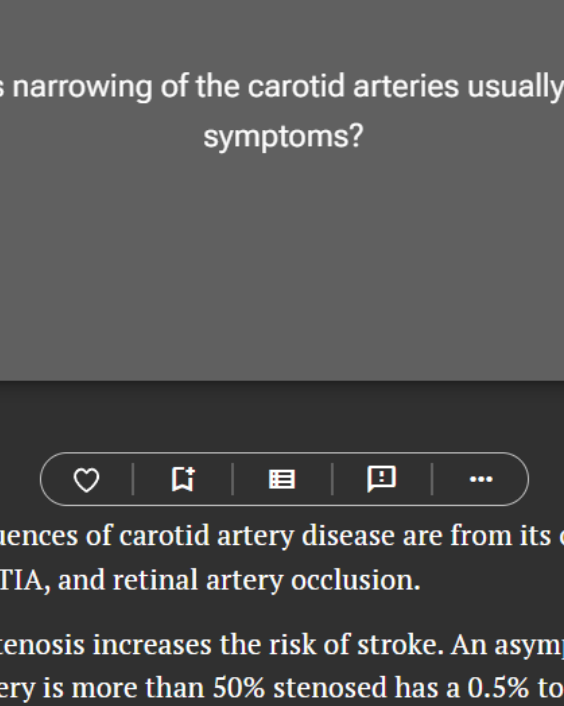

Figure 2. Embolic Ischemic Stroke.

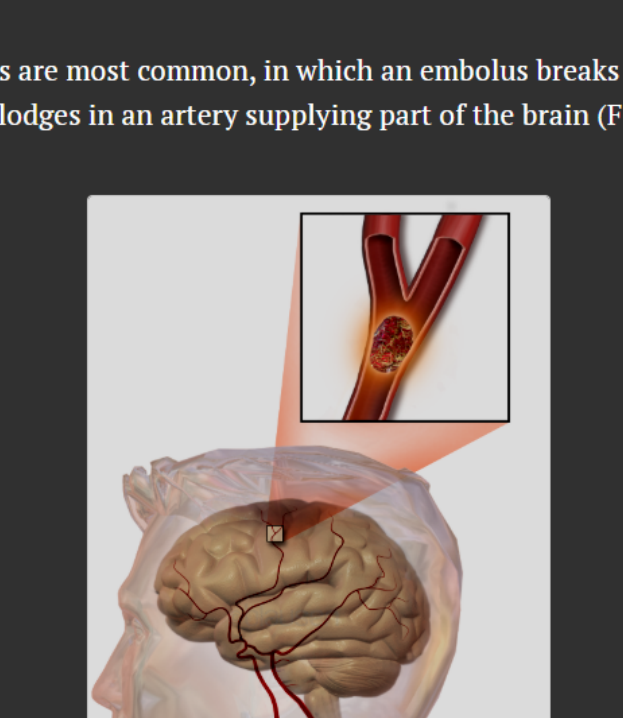

Figure 3. Right Middle Cerebral Artery Embolus. At the red arrow, the right middle cerebral artery is occluded by an embolus of carotid origin.

CLINICAL CORRELATION

Symptoms of the stroke depend on the area of the brain affected. A stroke involving the middle cerebral artery would cause weakness or paralysis on one side of the face and body, speech difficulties, and potential visual or sensory changes. The middle cerebral artery is the largest of the cerebral arteries and the most common arteri[illegible]

Transient Ischemic Attack. In addition to stroke (in which the neurologic deficits are persistent for >24 hours), carotid disease can also lead to TIA, the sudden onset of neurologic deficits that spontaneously resolve within 24 hours (often within 1-2 hours). TIAs may occur after an embolus dissolves on its own or dislodges, so symptoms last only for a short time. Although self-limited, TIA significantly increases the risk of an ischemic stroke, so prompt treatment is crucial. If the source of the embolus (the carotid artery) is not treated promptly, patients with TIA are prone to future emboli that may cause permanent neurologic damage.

Retinal Artery Occlusion. Carotid emboli may also lodge in the retinal arteries, leading to occlusion of blood flow to the eye and consequent retinal ischemia and blindness. The patient usually experiences temporary, painless, monocular loss of vision followed by complete recovery. They may report the feeling of a "curtain falling" in front of their eye. This temporary loss of vision is called **amaurosis fugax** (Greek for "fleeting darkness"). Less often, permanent painless loss of vision may result.

Retinal artery occlusion may be diagnosed on fundoscopy (Figure 4).

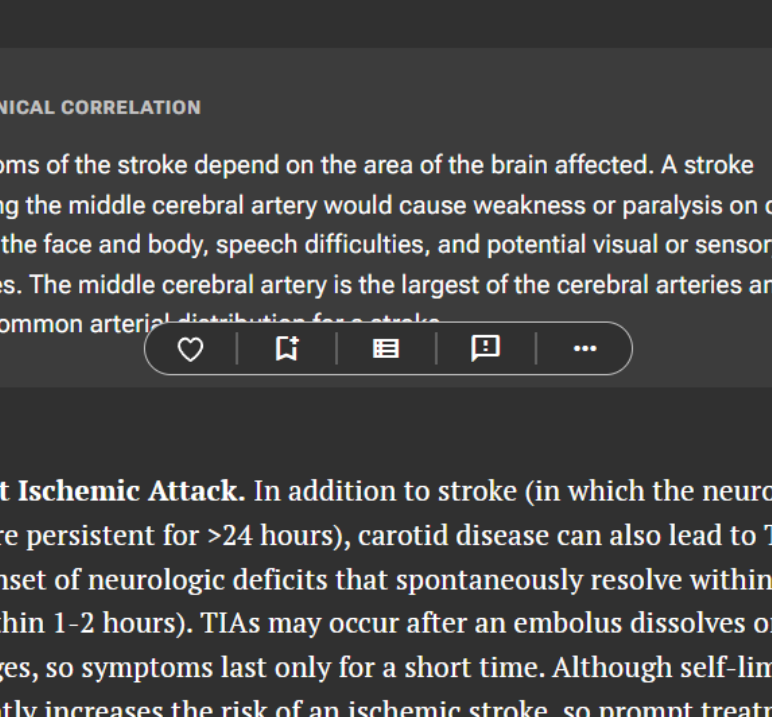

Figure 4. Normal Retina (left) versus Retina with Central Artery Occlusion (right). The two left-side images demonstrate a normal retina; take note of the normal appearance of the macula. Compare to the two right-side images, which demonstrate the features of central artery occlusion, namely a cherry-red appearance of the macula and the appearance of cotton wool spots.

Note the areas of ischemia (cotton-wool spots) and a cherry red spot in the central macula (AKA, the fovea) that are consistent with central retinal artery occlusion. The fovea is red because it is supplied by choroidal circulation, whereas the rest of the retina is pale because of obstructed flow in the retinal artery.

Amaurosis fugax carries a high risk of stroke within the following 3 months, so it must be addressed urgently to reduce the risk of stroke and of permanent and irreversible blindness.

What are some possible complications of carotid stenosis?

How Do We Diagnose Carotid Stenosis? (LO #3)

Carotid stenosis is diagnosed based on clinical and radiologic findings.

Physical Examination

Palpation and auscultation of carotid pulses are the most specific steps in the physical examination of a person with carotid stenosis.

Positive findings may include a **bruit** as well as absence of pulsation (however, the absence of one or both signs does not exclude the diagnosis). Note that auscultation should be done **before** palpation of the artery. Don't palpate if you hear a bruit, because palpation might dislodge a plaque from the carotid occlusion.

Imaging

Definitive diagnosis is made by imaging, which will assess the degree of stenosis to determin[illegible]id stenosis is classified into four categories based on the level of occlusion indicated on imaging:

- Mild disease: 0% to 49%
- Moderate disease: 50% to 79%
- Severe disease: 80% to 99%
- Occluded vessel: 100%

Initial testing can be done using carotid duplex ultrasound, MR angiography, or CT angiography, all of which have similar utility.

Carotid Duplex Ultrasound. This noninvasive test uses sound waves to estimate stenosis at the carotid bifurcation, primarily due to atherosclerotic plaque. Grayscale imaging is used to visualize the arterial plaque, and Doppler US calculates the degree of stenosis. Because the velocity of blood flow will increase at a narrowing, the Peak systolic Velocity (PSV) in the Internal Carotid Artery (ICA) is compared the PSV in the Common Carotid Artery (CCA), with increasingly elevated ICA/CCA PSV ratios consistent with increasing degrees of stenosis.

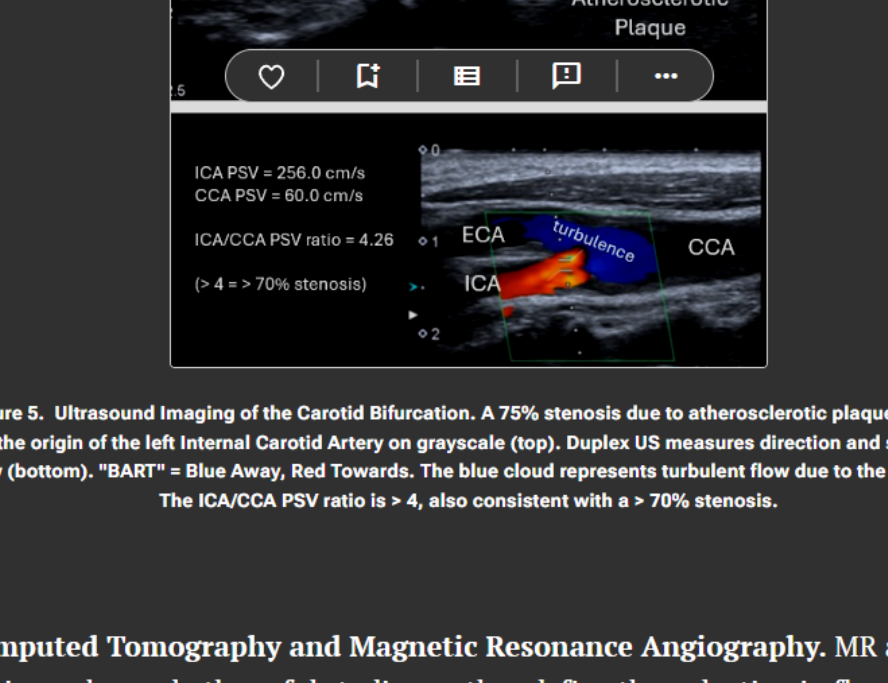

Figure 5. Ultrasound Imaging of the Carotid Bifurcation. A 75% stenosis due to atherosclerotic plaque is noted at the origin of the left Internal Carotid Artery on grayscale (top). Duplex US measures direction and speed of flow (bottom). "BART" = Blue Away, Red Towards. The blue cloud represents turbulent flow due to the stenosis. The ICA/CCA PSV ratio is > 4, also consistent with a > 70% stenosis.

Computed Tomography and Magnetic Resonance Angiography. MR and CT angiography are both useful studies, as they define the reduction in flow as well as the anatomic structure of the stenotic artery (Figure 6).

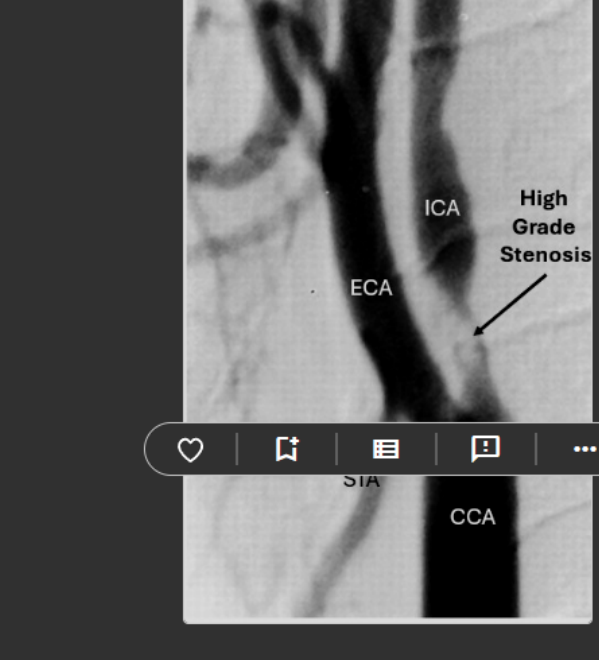

Figure 6. CT Angiogram. The Common Carotid Artery is seen bifurcating into the External Carotid Artery and Internal Carotid Artery. High grade ICA stenosis is noted at the carotid bifurcation. The ECA supplies the neck and face with arterial branches, starting with the Superior Thyroid Artery. The ICA has no branches until it enters the skull.

Although traditional carotid angiography is still used preoperatively, it requires an arterial catheter insertion for installation of contrast dye, and so it is not used in initial diagnosis.

Which are the best imaging studies to diagnose carotid stenosis?

How Do We Ma[illegible] #4)

Initial management of carotid stenosis for all patients is aimed at modifying risk factors to **lessen the development of atherosclerosis**, including lowering blood pressure and cholesterol levels via weight loss (if obese), instituting the Dietary Approaches to Stop Hypertension (DASH) diet and regular aerobic exercise, limiting alcohol consumption, and stopping smoking.

Medical Treatment

All patients with carotid stenosis (regardless of degree of occlusion) should receive intensive medical therapy designed to reduce thrombus formation and atherosclerosis. This includes:

- Statin drugs
- Aspirin (or clopidogrel if they cannot tolerate aspirin)
- Antihypertensive drugs to reduce BP to <120–130/80 mm Hg
- Strict glycemic control in patients with diabetes mellitus

Other associated disorders, such as myocardial infarction or stroke, may warrant additional medical therapy.

What is the basic treatment for a patient with asymptomatic carotid stenosis?

Surgical Management

Surgical management should be considered in patients with >70% stenosis, particularly if their life expectancy is >5 years and their perioperative risk is low. The surgical options are carotid artery angioplasty with placement of stents (Figure 7) or surgical endarterectomy (Figure 8), in which the thrombus is removed during an open surgical procedure.

Figure 7 is an angiogram of a right carotid artery both before (left) and after (right) angiography with stenting.

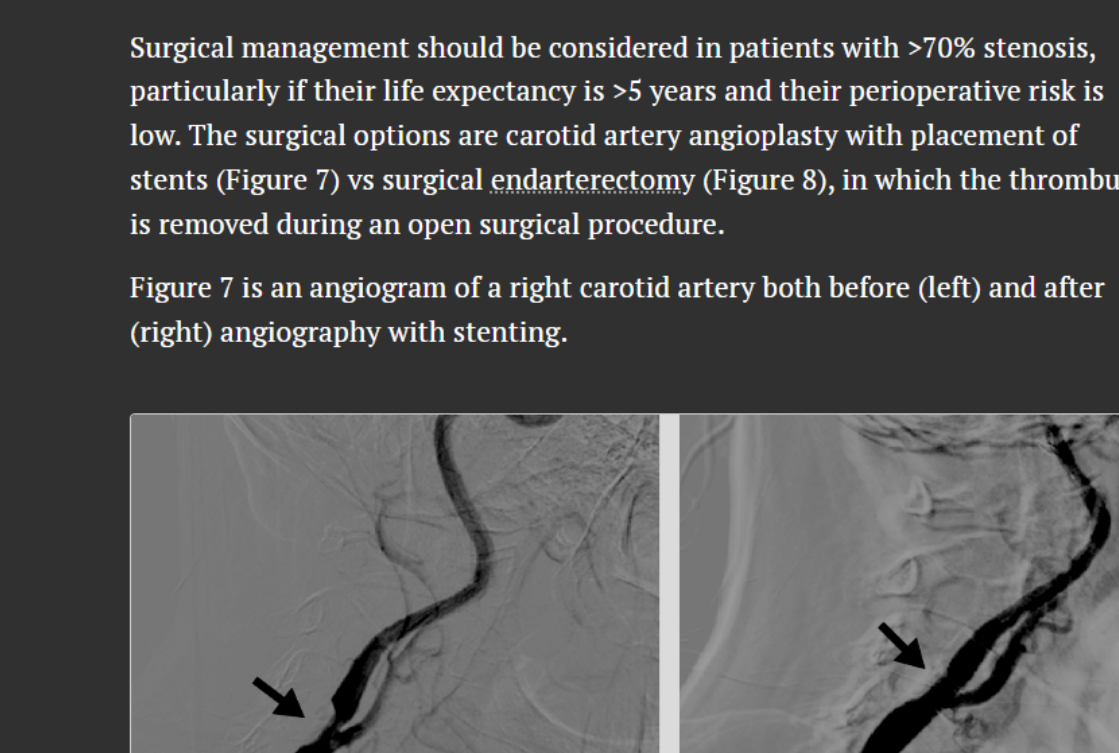

Figure 7. Right Carotid Artery Before and After Angioplasty.

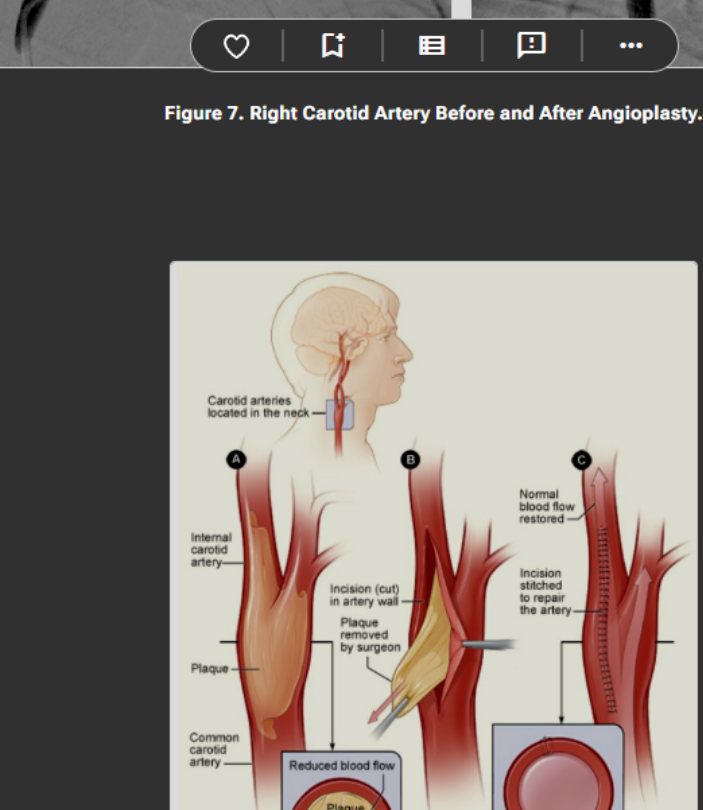
Carotid arteries located in the neck
A
B
C
Internal carotid artery
Incision (cut) in artery wall
Plaque removed by surgeon
Normal blood flow restored
Incision stitched to repair the artery
Plaque
Common carotid artery
Reduced blood flow
Plaque
Narrowed artery cross-section
Normal blood flow restored

Figure 8. Carotid Endarterectomy. Surgical removal of atherosclerotic plaque at the carotid bifurcation, with resolution of the stenosis.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TR, what is the likely diagnosis? What advice would you give her?

"It sounds like your doctor gave you good advice," you begin. You tell TR that her temporary loss of vision, called amaurosis fugax, was caused by a small plaque or clot in her carotid artery that broke off, traveled downstream, and lodged in the artery that provides blood flow to the eye. You explain that even though she is currently asymptomatic, she is at risk for an ischemic stroke that may result in permanent neurologic damage, including blindness. "I understand," TR says, "but can't I just take a pill? Do I really need surgery?" You explain that because of her recent symptoms and the severe blockage in her right carotid artery (80%-99%), surgery is the recommended treatment, in addition to aspirin therapy and control of her risk factors (hypertension, diabetes, hypercholesterolemia, and tobacco use). "OK, that's two votes for surgery, so I guess I'll do it," TR says. "I certainly don't want to have a stroke."

Summary

- Carotid stenosis is narrowing of the common and internal carotid artery caused by atherosclerosis.
- It is commonly found in older adult patients with atherosclerotic risk factors such as smoking, poorly controlled hypertension, hyperlipidemia, and hyperglycemia; carotid stenosis is usually asymptomatic.
- Complications such as stroke may lead to neurologic deficits.
- Thrombosis in the carotid artery leads to emboli which migrate to the cerebral or optic arteries.
- Severe stenosis narrows the lumen, compromising blood flow to the brain, but this is usually overcome by collateral flow through uninvolved arteries.
- Decreased blood flow to brain cells results in weakness, dizziness, and syncope.
- Complications ar[illegible]ebral arteries, leading to stroke or transient ischemic attack (TIA).
- Emboli may also lodge in the retinal artery or its branches and result in transient or permanent blindness.
- Physical examination may show a carotid bruit or reduced carotid pulses.
- Testing is done with doppler ultrasound, CT angiography, or MR angiography.
- Management involves lifestyle modifications such as weight loss, smoking cessation, and aerobic exercise.
- Medical therapy includes statins, aspirin, and antihypertensive and antidiabetic agents as required.
- Surgical treatment with angiography/stenting or endarterectomy should be considered in patients with >70% stenosis.

Review Questions

1. A 75-year-old male presents to the outpatient clinic for an annual physical examination. He takes enalapril, a statin, and metformin for long-standing hypertension and diabetes. He smokes one pack of cigarettes per day and drinks one to two beers each night. His body mass index is 30 kg/m^2. His blood pressure is 135/95 mm Hg, pulse is 80 beats per minute, respiratory rate is 18 breaths/min, and temperature is 98°F. Physical examination reveals a bruit over the left carotid artery. Ophthalmic examination and fundoscopy are unremarkable. Which of the following is the most appropriate next step in the evaluation of the patient?

A. ○ Carotid angiography
B. ● Carotid ultrasonography
C. ○ Ophthalmology consult
D. ○ Reassurance
E. ○ Temporal artery biopsy

Hide Explanation

The correct answer is carotid ultrasonography (B). The carotid bruit in the presence of risk factors for atherosclerosis (eg, advanced age, smoker, obesity) suggests carotid artery disease. In such a patient, Doppler ultrasound or CT/MR angiography is done to detect and determine the severity of carotid stenosis. Carotid angiography (A) is a more invasive option and is not used in initial diagnosis. Ophthalmology consult (C) does not take precedence over carotid Doppler, as the patient has more risk factors for circulatory causes of blindness than for ocular causes. Sole reassurance (D) is not appropriate, as carotid bruit predicts the possibility of carotid stenosis. Temporal artery biopsy (E) is required to screen for temporal arteritis, which is highly unlikely in this case, as it typically presents with headache and jaw claudication as well as amaurosis fugax.

2. A 72-year-old female presents to the clinic after experiencing an episode of transient vision loss in her right eye, which she likens to a curtain falling. She is otherwise in good health and is physically active. A physical exam shows a soft right carotid bruit. Ultrasonography shows 80% stenosis in the right carotid ar[illegible]otid artery. Which of the following is most appropriate for long-term management in this patient?

A. ○ Bilateral endarterectomy
B. ○ Left endarterectomy
C. ○ Reassurance
D. ● Right endarterectomy

Hide Explanation

The correct answer is right endarterectomy (D), as the female patient has presented with symptomatic right carotid stenosis of >70%. Surgical management is an option, especially when the patient has a >5-year life expectancy and is a low operative risk, as are suggested by her good health. Bilateral endarterectomy (A) and left endarterectomy (B) are incorrect, as the stenosis of the left carotid artery does not fulfill the indications of endarterectomy (ie, the left carotid is <70% stenosed). Amaurosis fugax is a major symptom of carotid stenosis that must be taken seriously, so reassurance (C) is not appropriate.

3. A 70-year-old male is found to have a carotid bruit, and imaging reveals 25% stenosis of both carotid arteries. He has hypertension treated with enalapril. His blood pressure is 135/95 mm Hg. Which of the following is the best next step in management, in addition to lifestyle modifications to reduce atheroscle[illegible]

A. ○ Carotid angioplasty with stenting
B. ○ Carotid endarterectomy
C. ○ Statin and aspirin
D. ● Statin, aspirin, and increased hypertension control
E. ○ Statin, aspirin, clopidogrel, and increased hypertension control

Hide Explanation

The correct answer is statin, aspirin, and increased hypertension control (D). These reduce the risk of thrombosis, and BP should be controlled to <120-130/80 mm Hg in high-risk patients. Surgical management with carotid angioplasty with stenting (A) or carotid endarterectomy (B) is only used when patients have severe stenosis (>70%). Additional hypertension control is needed, because the BP is >130/80 mm Hg, so statin and aspirin (C) are insufficient. Adding a second antiplatelet drug increases side effects and does not clearly benefit the patient, so statin, aspirin, clopidogrel, and increased hypertension control (E) is incorrect.

4. Which of the following provides the best description for the pathogenesis of amaurosis fugax?

A. ○ A thrombotic event, in which rupture of a carotid artery atherosclerotic plaque leads to thrombus formation.
B. ○ A thromb[illegible]atherosclerotic plaque leads to thrombus formation.
C. ● An embolic event, in which the emboli originates from a carotid artery and lodges in the retinal artery.
D. ○ An embolic event, in which the emboli originates from a coronary artery and lodges in the carotid artery.

Hide Explanation

The correct answer is (C) - amaurosis fugax refers to a temporary, painless loss of vision that is caused when an embolus travels from a carotid artery plaque to occlude a retinal artery. Atherosclerotic carotid artery plaques can indeed rupture and induce overlying thrombus formation (A), but amaurosis fugax is specifically an embolic event involving the retinal artery. Likewise, answers (B) and (D) do not correctly characterize the pathogenesis of amaurosis fugax.

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of carotid stenosis.~~

- Walker BR, Colledge NR, Ralston SH, Penman ID, eds. *Davidson's Principles and Practice of Medicine.* 22nd ed. Churchill Livingstone; 2014.
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of carotid stenosis.~~

02 ☑ ~~Explain the pathophysiology of carotid stenosis.~~

03 ☑ ~~Interpret clinical findings to properly diagnose carotid stenosis.~~

04 ☑ ~~Outline the management plan of carotid stenosis.~~

Objective Confidence: 4/4